SUNY Old Westbury **2026-2027 Undergraduate Catalog**

Management Information Systems, B.S.

HEGIS Number: 0702 Curriculum Code: U0858 Management information systems (MIS) is a general term used to describe the computer-based information systems that provide operational information to an enterprise. The SABRE System is an example of a computer-based information system. Originally used only by American Airlines for passenger reservations, it is today a system connecting more than 40,000 travel agents and millions of consumers with more than 400 airlines, 50 car-rental companies, thousands of hotels, railways, tour companies, ferries and cruise lines. The MIS term is often used to describe the roles who work as varies management and data analysis work in these computer-based information systems. The MIS degree program prepares students as technical managers who can rise to serve in various capacities in an enterprise, starting as System Analysts in MIS departments. The goal of the MIS program is to give students both a firm grounding in information processing as well as in business fundamentals. Therefore, MIS majors take a broad range of required courses both in CIS and business. The MIS curriculum includes courses in database management, software engineering, Internet and Web Technology, system design & implementation, and technical communications. These courses provide MIS majors with an extensive background in technical and organizational areas which serves as a solid foundation for understanding how new technological advances enhance the use of information resources for operations and strategic decision-making. Mission Statement The Management Information Systems (MIS) degree program prepares students as technical managers who can rise to serve in various capacities in an enterprise, starting as System Analysts in MIS departments. The goal of the MIS program is to give students both a firm grounding in information processing as well as in business fundamentals. MIS students graduate with an extensive background in technical and organizational areas which serves as a solid foundation for understanding how new technological advances enhance the use of information resources for operations and strategic decision making. Learning Outcomes Problem Solving through iterative design and algorithmic thinking including Problem Formulation through Abstraction. Expressing solutions Computationally. Analysis and Evaluation of Solution Execution including memory and runtime efficiency. Principles of object-oriented software engineering as applied to the development of large scale software sys tems in a team environment. High level programming languages as related to database application programming in business problem solving. Understanding of the the internet and developing networked applications.

Requirements for the Bachelor of Science (B.S.) Degree in Management Information Systems

Major Course Requirements: 66 Credits (17 courses)

1. Computer Science Courses: 36 Credits (9 courses)

- CS 2510 - Computer Programming I Credits: 4
- CS 2511 - Computer Programming II Credits: 4
- CS 3611 - Advanced C# Credits: 4
- CS 4100 - Technical Communications Credits: 4
- CS 4501 - Software Engineering Credits: 4
- CS 4550 - Database Management Systems Credits: 4
- CS 4720 - Internet and Web Technologies Credits: 4
- CS 5710 - Computer Networks Credits: 4
- CS 5910 - System Design & Implementation Credits: 4

3. Business Courses: 14 Credits (4 courses)

- BU 3502 - Principles of Accounting I Credits: 3
- BU 3511 - Principles of Accounting II Credits: 3
- BU 3800 - Marketing: Principles and Concepts Credits: 4
- BU 4110 - Organizational Behavior and Management Credits: 4

4. Economics & Analytics Courses: 8 Credits

(2 courses)

- MA 2701 - Business Anaytics Credits: 4
- PE 2420 - Principles of Microeconomics Credits: 4
- PE 2430 - Principles of Macroeconomics Credits: 4

5. Mathematics Courses (Cognates): 8 Credits

(2 courses)

4a. One of the following:

- MA 2300 - Calculus for Business and Economics Credits: 4
- MA 2310 - Calculus & Analytic Geometry I Credits: 4
- MA 3030 - Discrete Mathematics Credits: 4

4b. One of the following:

- MA 2000 - Applied Statistics Credits: 4
- MA 3210 - Introduction to Probability and Statistics Credits: 4

Liberal Education Curriculum Courses

Refer to the Liberal Education Curriculum Bulletin for requirements.

General Elective Courses

In consultation with academic advisor, for a total of 120 credits.

Other Requirements

- A grade of C or higher is needed in all required computer science, business, economics, and mathematics courses.
- Transfer students must complete a minimum of 28 credits (7 courses) of the required major courses at or above the 3000 level at Old Westbury.

College Wide Requirement

- 120 credits overall (40 credits at Old Westbury, may transfer a maximum of 80 credits)
- 45 Upper Division credits (3000, 4000, or 5000 level courses)
- 60 Liberal Arts credits
- Cumulative Grade Point Average of 2.0

Sample Course Sequence By Major

Term 1

- PE 2420 - Principles of Microeconomics Credits: 4
- MA 2080 - Precalculus for Business and Economics Credits: 4

Term 2

- PE 2430 - Principles of Macroeconomics Credits: 4
- CS 2510 - Computer Programming I Credits: 4

Term 3

- CS 2511 - Computer Programming II Credits: 4
- BU 3502 - Principles of Accounting I Credits: 3
- MA 3030 - Discrete Mathematics Credits: 4

  OR
- MA 2300 - Calculus for Business and Economics Credits: 4

Term 4

- CS 3611 - Advanced C# Credits: 4
- BU 3511 - Principles of Accounting II Credits: 3
- MA 2000 - Applied Statistics Credits: 4

  OR
- MA 3210 - Introduction to Probability and Statistics Credits: 4

Term 5

- CS 4501 - Software Engineering Credits: 4
- CS 4550 - Database Management Systems Credits: 4
- BU 3800 - Marketing: Principles and Concepts Credits: 4

Term 6

- CS 4100 - Technical Communications Credits: 4
- CS 4720 - Internet and Web Technologies Credits: 4

Term 7

- CS 5710 - Computer Networks Credits: 4
- BU 4110 - Organizational Behavior and Management Credits: 4

Term 8

- CS 5910 - System Design & Implementation Credits: 4